

Machine Learning-based SEM Images Contour Extraction Data Augmentation

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ABSTRACT

Scanning Electron Microscopy (SEM) images contour extraction is crucial due to its ability to provide precise surface morphology analysis. Accurate contour extraction is helpful for quality control, failure analysis, defect detection and the development of advanced devices. However, existing machine learning-based contour extraction methods face the problem of difficulty in obtaining enough training datasets, particularly the lack of effective SEM image/contour pairs, resulting in defects in the extracted contours, such as open line ends and discontinuous lines.

In this work, we propose a cGAN-based data augmentation pipeline to solve these issues. We generate defective contours by artificially introducing defects into ideal contours, which then serve as input to a cGAN-based model to generate synthetic SEM images with low contrast or weak signal in the corresponding parts. These generated SEM images, paired with original ideal contours, are then used to train a new contour extraction model, which significantly improve the model's ability to handle previous failed cases. The experimental results demonstrate that our approach greatly enhances the contour accuracy, resolving previously challenging cases.

Keywords: SEM images, contour extraction, conditional GAN, pix2pix, data augmentation, image generative model

1. INTRODUCTION

SEM images contour extraction is a crucial task in semiconductor manufacturing, where precise morphology analysis is important for quality control and defect detection. However, the lack of sufficient SEM image/contour training pairs limits the performance of our existing machine learning models, leading to common defects such as unclosed line ends and discontinuous lines, particularly in low-contrast or weak signal regions.

Our previous work^[1] implemented a machine learning-based contour extraction model that demonstrated promising results but faced challenges in some cases, where the model failed to extract closed line ends or produced discontinuous contours. A comprehensive dataset with SEM images paired with the corresponding correct contours would be ideal for improving the model performance, but such data is hard to obtain.

To overcome this issue, we propose a data augmentation pipeline using a cGAN-based generative model, as illustrated in Figure 1. The process begins with contour extraction using our previous contour extraction model. These extracted contours, which contain defects like open line ends, are paired with their corresponding SEM images to train a cGAN model(pix2pix). We then create high-quality contours (C1) using OPCV. Artificial defects are introduced into C1 to generate a defective version with specific types of defects(C2), which serves as input to the cGAN model. The model generates synthetic SEM images corresponding to these defective contours, mimicking weak signal or low contrast parts. The generated synthetic SEM images are then paired with the ideal contours(C1) to form an augmented training dataset for the contour extraction model.

By incorporating this new synthetic SEM images dataset, our approach significantly enhances the ability of the contour extraction model to handle the previously challenging cases. The newly trained model demonstrates improvements in closing line ends and reducing discontinuities, addressing the problems observed in the previous model.

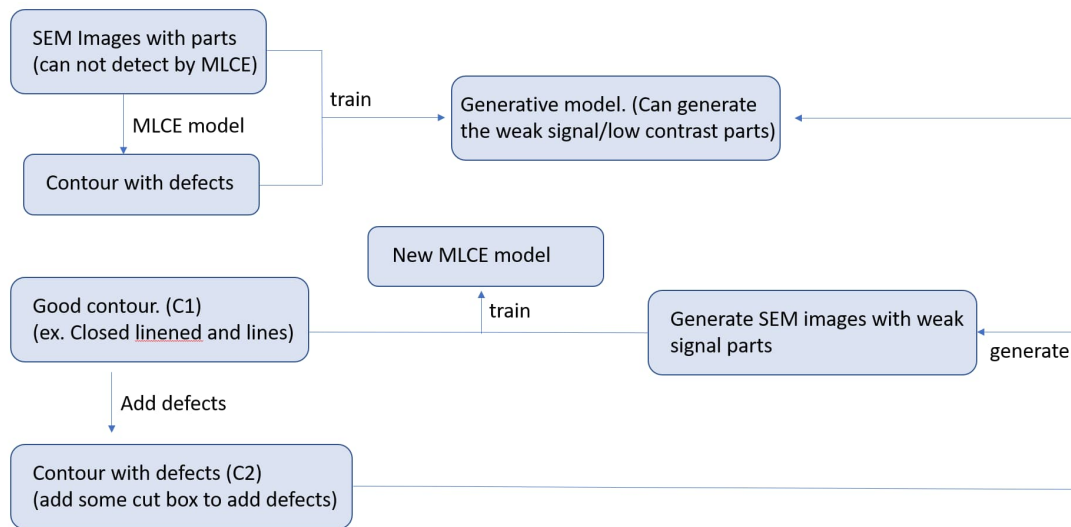


Figure 1. Overview of the cGAN-based data augmentation pipeline for SEM contour extraction

2. CGAN-BASED DATA AUGMENTATION FRAMEWORK

Challenges in Contour Extraction

Our previous contour extraction model encountered difficulties in some regions. One of the most common defects in our previous work was open line ends, as illustrated in Figure 2. These defects result in incomplete contour extraction, making further analysis and applications unreliable.

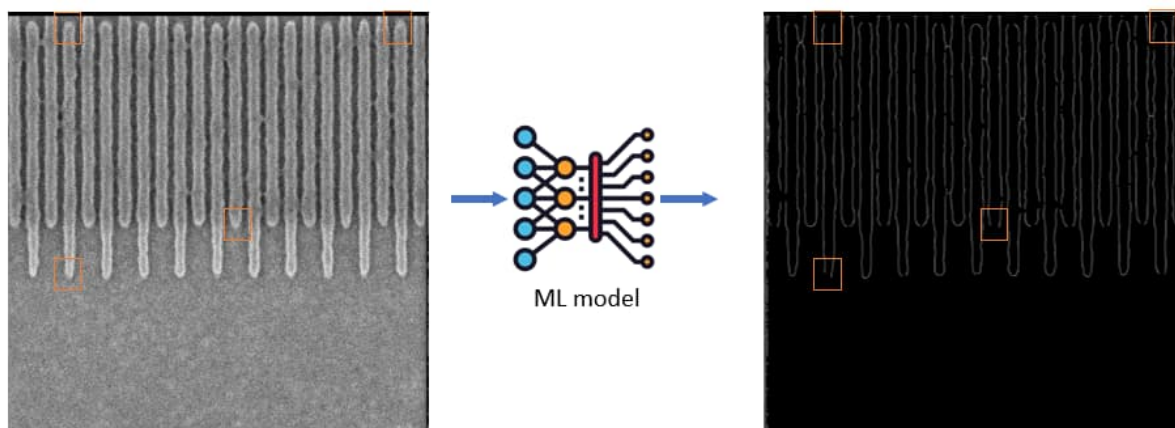


Figure 2. Failure case of the original contour extraction model with open line ends.

To address these issues, we use the extracted defective contours paired with their corresponding SEM images to train a generative model. Several generative models were considered, including cGAN-based Pix2Pix^[2], latent diffusion models^[3], and other deep-learning generative approaches. After evaluating these models, we selected Pix2Pix, which

performs better on small datasets. Since we have limited number of SEM image and contour pairs, the cGAN-based model is the most suitable one for our task.

Model Training

The cGAN based Pix2Pix model learns a mapping from input images (contours with defects) to output images (SEM images). Unlike standard GAN models, which generate images from random noise, Pix2Pix is conditioned on input images, which make it particularly suitable for specific image-to-image translation tasks^[2].

A cGAN consists of two neural networks: a generator and a discriminator. The generator aims to generate a realistic SEM image with an input contour image, while the discriminator attempts to distinguish between real SEM images and synthetic SEM images. The two networks are trained adversarial where the generator tries to improve its outputs to fool the discriminator, and the discriminator refines its ability to differentiate real from generated images. This adversarial training process ensures that the generated SEM images closely resemble the real SEM images.

One key component of the Pix2Pix framework is the Patch-GAN discriminator structure, which classifies whether small patches (instead of the entire image) are real or fake. This approach makes the model have better local details in generated SEM images. The original Pix2Pix implementation uses a 70×70 Patch-GAN^[2]. However, for our 1024×1024 images cases, this configuration led to unwanted periodic repetition of small patterns in SEM image backgrounds, as shown in Figure 3(a)(b).

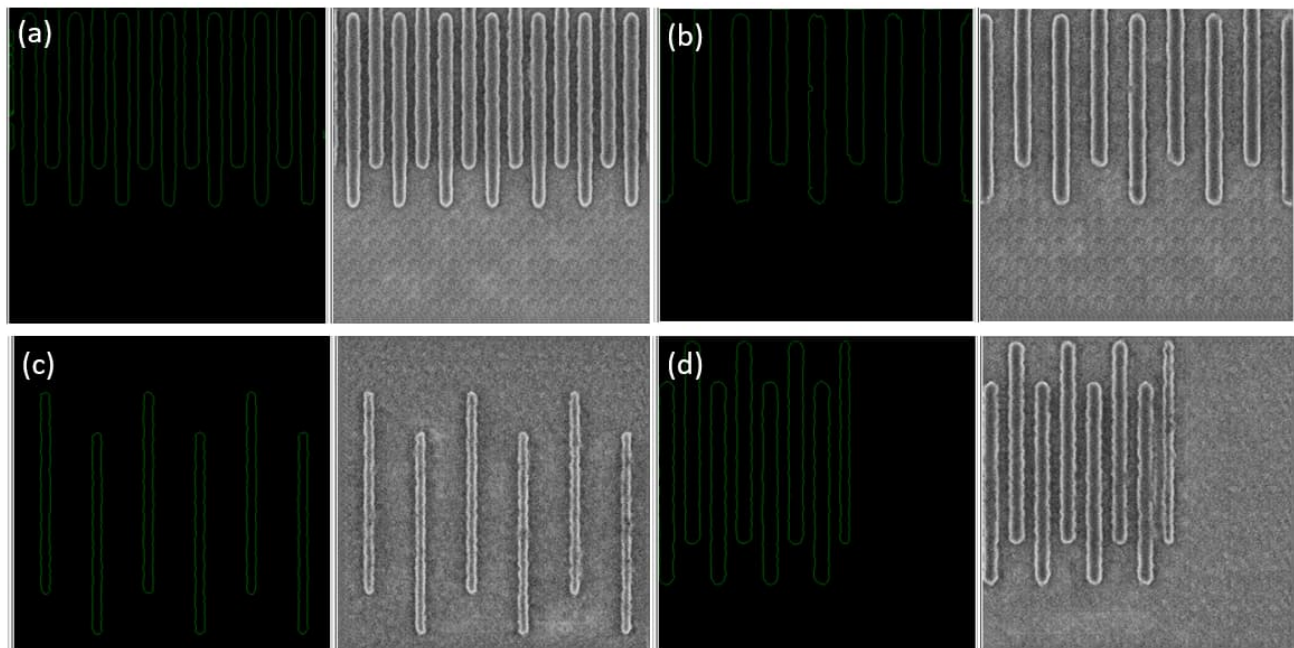


Figure 3. Sample results of synthetic SEM images. (a),(b) results with small patch-GAN showing periodic background artifacts. (c),(d) results with large patch-GAN showing improved result.

To solve this issue, we increased the Patch-GAN receptive field. This adjustment significantly improved the quality of the generated SEM images, eliminating those repeating patterns in the background while maintaining fine details of the structures.

We trained the Pix2Pix model using 360 pairs of 1024×1024 SEM images and their extracted defective contours. Figure 3 shows the sample results of different Patch-GAN settings:

(a), (b) Small Patch-GAN leads to periodic patterns in the background.

(c), (d) Larger Patch-GAN improves generated results, eliminating repeating patterns and enhancing realism.

Contour Processing: Generating C2 from C1

To generate defective contours (C2) from ideal contours (C1), we developed a defect injection algorithm. This method identifies line-end regions in C1 and selectively removes a part of the contour to introduce defects. In our case, 80% of line-end parts are selected. Then a box with a random size of 8–20 pixels will be applied to remove segments at those selected parts.

This process generates the defective contours (C2), which are then used as inputs of the previously trained generative model to generate synthetic SEM images containing weak signal or low contrast regions. Figure 4 and 5 illustrates this process.

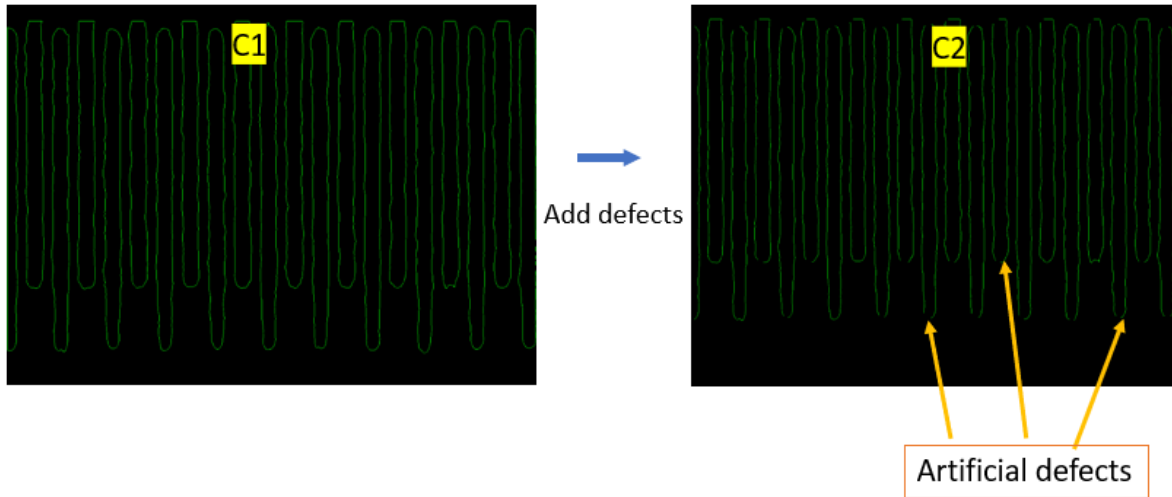


Figure 4. Process of introducing defects to contours(C1 to C2).

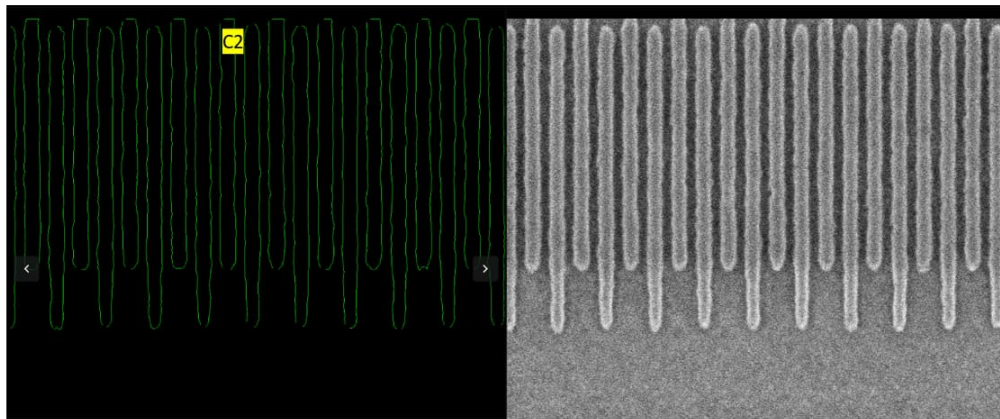


Figure 5. Generation of synthetic SEM images using defective contours(C2).

Training the New Contour Extraction Model

After generating synthetic SEM images, we pair them with their corresponding ideal contours (C1). This newly created dataset is then used to train a new contour extraction model. With the augmented data, the new model can learn to better handle the weak signal or low contrast regions and correctly extract contours, particularly in those challenging cases

where the original model failed. The improved model is expected to show enhanced performance in detecting contours with closed line ends.

3. MODEL ENHANCEMENT FOR LINE-END COMPLETION IN SEM CONTOURS

To evaluate the effectiveness of our proposed pipeline, we compared the performance of the newly trained contour extraction model with the original model. The primary focus was on the improvement in handling open line ends, a major issue encountered in our previous work.

Our results demonstrate that the augmented dataset significantly enhanced the model's ability to extract contours with closed line ends. As shown in Figure 6, the original model failed to extract closed line ends in some regions. In contrast, the new model trained on the augmented dataset, successfully resolved most of open line-ends, confirming the feasibility of our approach.

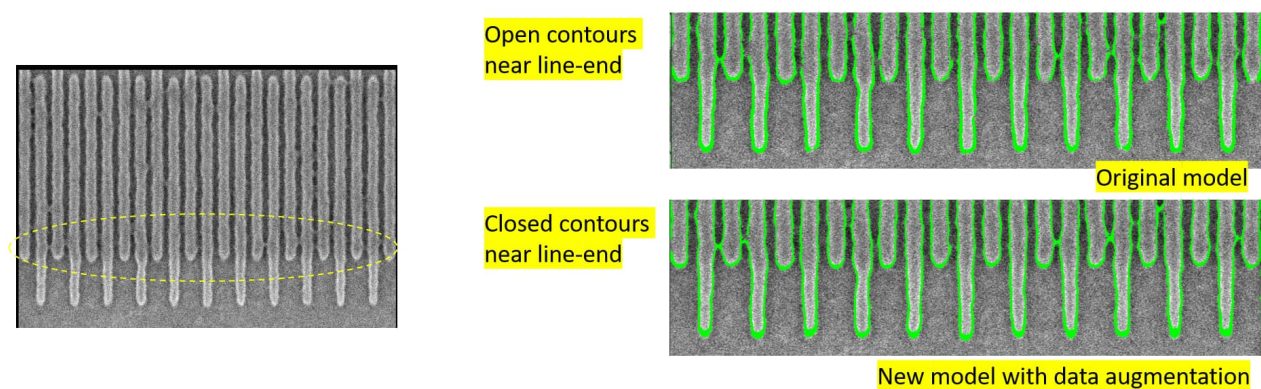


Figure 6. Comparison of contour extraction results between the original model and new model with data augmentation.

Figure 6 illustrates a visual comparison, the result above shows original model's contour extraction results, where open line ends are apparent and the result below shows the new model's output, showing closed line-end parts.

The comparison highlights the advantage of using our data augmentation pipeline. By generating synthetic SEM images with weak signal or low contrast regions and pairing them with high-quality contours, our method effectively provided the contour extraction model with additional learning signals, leading to more accurate contour extraction.

4. CONCLUSIONS AND FUTURE WORK

In this work, we proposed a data augmentation approach to improve SEM image contour extraction, particularly focusing on addressing the issue of open line ends. By generating synthetic SEM images with weak signal or low contrast conditions with generative model and pairing them with ideal contours, our method successfully enhanced the performance of our contour extraction model. The experimental results demonstrated that our approach effectively solved most of previously open line ends cases, showing the effectiveness of using the data augmentation pipeline in this task.

Beyond addressing open line-end defects, our proposed pipeline is generalizable to improve contour extraction with other defects. The method of introducing controlled defects and using a generative model to simulate the weak signal or low contrast conditions can be extended to various defects, such as discontinues lines. This demonstrates the broader application of our approach to enhancing SEM image analysis.

The proposed method highlights the potential of image generative models in data augmentation of limited training datasets to improve performance in scientific imaging applications. However, some areas remain open for further exploration. Beyond Pix2Pix, we can also explore alternative architectures such as latent diffusion models or other deep-learning approaches, which might provide further improvements in the quality of generated SEM images. But these methods require larger dataset. Since our study was now performed on a small dataset of 360 paired 1024*1024 SEM images and contours, extending the data augmentation pipeline to larger datasets could provide more robust and comprehensive results.

Overall, this work demonstrates the feasibility of using image generative models for SEM images contour extraction data augmentation, paving the way for more robust contour extraction techniques in microscopy imaging.

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